Day: 2025(S)-CS-26 Date:

Problem #1

* Printing buterfly patter on console using *

Taking size in n;

* loop for counting stars

```
for (int row=1; row <= 2*n-1; row++)
  { int stars
      if row<=n
          then star=row
      else star=2*n-row
```

* for space: int space=2*(n-star)

* Printing star at j=1

```
//left   loop (j=1; j<=star; j++) << *
// righ   for(j=1; j< star j++  << *
//Gap
        for(j=1; j< space j++) << i
```

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Dry run

Stars Calculation suppose n=3 then according our code

	Star Calc	Stars	Space = 2x(3-stars)
i)	row≤3 → star=row=1	1	4
ii)	row≤3 → star=2	2	2
iii	row≤3 → star=3	3	0
iv	row>3 → star=6-4=2	2	2
v	row>3 → star=1	1	4

Line

1

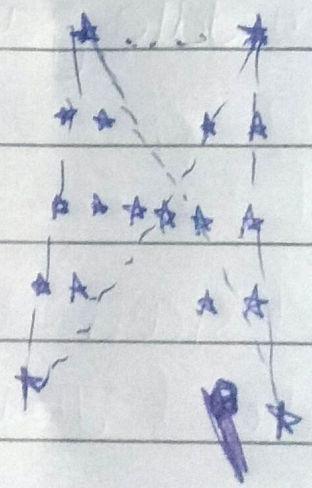
2

3

4

5

Output



y: Date:

Problem 2

printing

1	2	3	4	5
14	15	16	17	6
13	20	19	18	7
12	11	10	9	8

first 2D array with 4 rows & 5 coloums arr [4] [5]

A counter variable

int value = 1

storage in 2D array will be using this variable

for 1st row

```
for (int j=0; j<5; j++)
  { arr [0][j] = value++;
  }
```

it will start for value=1 to 5 and fill row 1 denoted by arr[0]

* Another loop to fill last col as

```
for (int i=1; i<3; i++)
{ arr[i][4] = value++;}
```

Now value start from 6 (Previou and store at arr[i][4] = [1][4] = 6

then 7 at [2][4] ++ will increment

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* Now last row

```
for (int j=4; j>0; j--)
{ arr [3][j] = value++
```

arr [3][4] = 8
arr [3] [3] = 9
arr [3][2] = 10
So on decrement
unitll = 12

after this value = 13 (store)

* for (int i=2; i>1; i--)
  { arr [i][0] = value++; }

arr [2][0] = 13
arr [1][0] = 14

Moving right arr value = 15

* for (int j=1; j<4; j++)
  { arr [1][j] = value++ }

arr [1] [1] = 15
arr [1] [2] = 16 So on till 17

now value = 18

* for (int j=3; j>0; j--)
  { arr [2][j] = value++ } arr[2][3] = 18
  so on to 20

After the will Print array using nested loop like matrix arr[i] [j]